



Figure 6: Segmented image

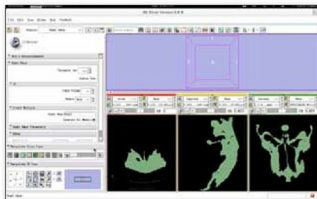


Figure 7: Model maker module

parameter label to 1 for tissue, as done earlier. Enter the name of the scene saved previously in the *Input Volume box*.

Now you are done! Hit *Apply* and wait for a few minutes while Slicer creates the 3D model of the scanned segmented image. You can zoom in or out by scrolling the mouse, and even rotate around the three axes. You can then save your model in the VTK format.

Applications

Imaging and analysis

Slicer integrates several facets of scanned image, and provides automatic registration, i.e., aligning of data sets or semi-automatic segmentation by which you can analyse the body structure for tumours and vessels. This part has been demonstrated in the above section. You can do qualitative analysis of various 3D models of the scanned image, like the one generated above, which includes measuring angles, distances, surface areas and volumes.

Diffusion Tensor Imaging (DTI)

When neural axons of white matter in the brain or muscle fibres in the heart acquire an internal fibrous structure, leading to unbalanced diffusion of water through the tissue, DTI is done. Simply put, DTI is a Magnetic Resonance Imaging technique that enables the measurement of restricted water in the tissue. 3D Slicer can take the DTI data and perform the analysis in a few steps.

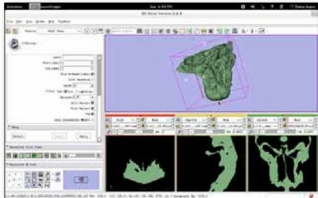


Figure 8: 3D model of skull

Neurosurgical planning

This deals with the generation of fibre tracts in the vicinity of a tumour. Here we export the same DTI scanned image and explore it with Slicer to perform neurosurgical planning.

ROI seeding

This is a tractography implementation that allows a user to seed tracts from a region of interest (ROI). The ROI is defined as a labelmap and has to be provided by the user. By using the editor module of Slicer described above, this can be implemented easily.

Biopsies

A recent application includes the robot-assisted MRI-guided prostate biopsy using 3D Slicer. This technique calibrates a robot to the MR coordinate system (as a digital device, a robot is ideally suited to accurately align an instrument to any point in the three-dimensional coordinate system of the magnetic resonance (MR) image) to guide it through the targeting volume and collect the scanned data to locate the tumour. The link below gives a brief overview of this magnificent technique, which interfaces with the robot through 3D Slicer: http://www.slicer.org/slicerWiki/images/0/06/ProstateNav_TutorialContestSummer2010.pdf

B-spline registration

B-spline registration in Slicer 3.6 allows for non-rigid registration between pre-procedure MRI and intra-procedure CT images during CT-guided tumour ablation in the liver. It provides increased tumour visualisation during the planning, targeting and monitoring phases of the ablation procedure.

Instrument tracking

Conventional image-guided surgery systems present the surgeon with data that was gathered prior to surgery, apart from tracking surgical instruments within the operating field and rendering the tracked devices along with the data. As a tracked surgical instrument is moved within the surgical field, it is rendered in the 3D view, and the reformatted slice planes follow its position, sweeping through the volumes as